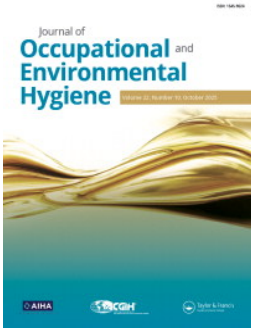

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Characterization of silver biocides and composites in commercial face masks with implications for risk assessment

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ABSTRACT

Silver-based biocides applied in fabric-based mouth- and nose-covering face masks require characterization due to the potential toxic effects of silver to which users may be exposed. In the absence of reliable silver release data in realistic usage conditions, current safety assessment of face masks relies on a safe-by-design principle. To contribute to the refinement of specifications of safe face masks, types of particulate silver biocides actively applied in face masks on the market were identified and characterized. Ultramicrotomy followed by scanning transmission electron microscopy coupled with energy-dispersive X-ray spectroscopy (STEM-EDX) analysis showed that a wide variety of silver-based biocides were applied. This includes metallic silver nanoparticles (NPs), silver salts in NP form, silver ion exchangers, and notably, various silver nanocomposites with other particulate materials such as synthetic amorphous silica (SAS), TiO_2, and ZnO. These composites are added to face masks to combine different modes of biocidal action or to facilitate gradual silver release, thereby extending biocidal activity. In five out of seven silver-containing masks, silver-containing NPs were identified on the surface of fibers. Additionally, significant numbers of other NPs (SAS, TiO_2) were found coating the fiber surfaces. Sizes of silver-containing NPs ranged from 3 to 200 nm across all masks, with the large majority of particles below 50 nm. These findings imply that for safety assessment, no-adverse effect levels of all incorporated compounds should be taken into account and that the effects of co-exposure to multiple compounds should be considered. The completion of particle release studies, exposure assessments, and regulatory oversight of face masks is recommended.

KEYWORDS
Electron microscopy; fiber; nanoparticles; nanomaterials; ultramicrotomy

Introduction

Silver (Ag)-based biocides are implemented into fabric-based mouth- and nose-covering face masks to provide antimicrobial properties for increased protection of the user (Li et al. 2006; Blevens et al. 2021; Abazari et al. 2023). This application expanded notably during the COVID-19 pandemic to various types of face masks, including cloth, surgical, and respiratory masks (Mast et al. 2023). Silver can either be added to textile fibers as a finishing coating by methods such as impregnation (Mahmud et al. 2017; Valdez-Salas et al. 2021; Caldas et al. 2024), chemical *in-situ* formation (Montes-Hernandez et al. 2021; Abazari et al. 2023), and spray coating (Sataev et al. 2014; Trabucco et al. 2021) or incorporated into fibers before extrusion (Yeo et al. 2003) or by electrospinning (Hong et al. 2006; Zhang et al. 2007) for synthetic polymer fibers. In the latter case, Ag^+ ions are gradually released and diffuse to the surface to facilitate increased biocidal activity (Gao and Cranston 2008).

The three main categories of commercial Ag biocides application in textiles are metallic Ag, Ag salts, and Ag ion exchangers (Gao and Cranston 2008; Windler et al. 2013; Zanoaga and Tanasa 2014; Simoncic and Klemencic 2015). Ag nanoparticles (NPs) are the most frequently reported within the metallic Ag category (Zhang et al. 2009; Hebeish et al. 2011; Radetić 2013), while other formulations include Ag particles combined with other compounds, such as

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Ag NPs on a silica core (Nischala et al. 2011) or a Ag-titanium dioxide (TiO_2) nanocomposite (Chacon-Argaez et al. 2023). Textiles functionalized with Ag salts, such as silver chloride (AgCl) NPs on silk (Potiyaraj et al. 2007) and AgCl embedded in a silica matrix applied to cotton (Tomšič et al. 2009) are reported. Additionally, Ag ion-based exchangers applied in textiles include materials like Ag zeolites (Dutta and Wang 2019) (e.g., Agion) or zirconium phosphates and glasses embedded with Ag^+ ions (e.g., Alphasan, SilverShield) (Morais et al. 2016).

A similar variety of Ag-based biocides applied in textiles has also been reported in face masks. Scientific proof-of-concept studies demonstrating antibacterial and virucidal efficacy of Ag-treated masks have primarily focused on Ag nanoparticle (NP) coatings on fibers. Valdez-Salas et al. (2021) demonstrated controlled and well-distributed impregnation of Ag NPs (average size 13.2 nm) on surgical masks via a nano-disinfectant application. Abazari et al. (2023) sonochemically coated Ag NPs (average 100–125 nm) onto polypropylene fabrics for face masks, showing that increased sonication time and precursor concentration enhances deposition and coating stability. Ferrari et al. (2024) employed the Ag mirror reaction to densely coat the polypropylene outer layer of masks with Ag NPs averaging 150 nm. In all cases, Ag NPs were physically bonded to the fibers, as confirmed by the absence of chemical modifications in fiber properties. Others demonstrate the application of Ag nanocomposites in combination with silica as sputter coating on a FFP3 mask (Balagna et al. 2020) or with TiO_2 added as a second antimicrobial compound (Li et al. 2006). Studies examining a broader range of Ag-treated face masks additionally report application of ionic Ag^+ (Mast et al. 2023), polycationic polymers that bind Ag^+ ions (such as those employing Silvadur technology) (Mast et al. 2023), and Ag ion-based exchangers like zeolites (Blevens et al. 2021).

Potential adverse effects of Ag exposure, whether in ionic or NP form, raise safety concerns about the use of Ag-treated masks. *In vivo* studies indicate that Ag NP exposure through inhalation, ingestion, or skin contact may cause local inflammatory responses, oxidative stress, and translocation into the bloodstream, leading to accumulation and toxicity in various organs (Takenaka et al. 2001; Gaillet and Rouanet 2015; Hadrup et al. 2018; Ferdous and Nemmar 2020). *In vitro* studies have shown cellular uptake of Ag NPs across different cell lines, causing cell shrinkage, cell death, membrane or DNA damage, and reduced cell viability (Carlson et al. 2008; Kim and Ryu 2013; Ferdous and Nemmar 2020). These cyto- and genotoxic risks depend on the NP's properties such as size, shape, and surface properties, as well as dose and duration of exposure (Butler et al. 2015; Abdal Dayem et al. 2017; Ferdous and Nemmar 2020). Oral toxicity studies (Hadrup and Lam 2014) and genotoxicity and cytotoxicity (Guo et al. 2016; Li et al. 2017) studies report similar dose-dependent effects for Ag NPs and Ag^+ ions. Uncertainties remain whether toxic effects stem directly from the NPs themselves or from the release of Ag^+ ions, and further research is needed to clarify the mechanisms of action.

Conducting a comprehensive risk assessment for application of Ag-based biocides in face masks remains challenging due to limited data on Ag release from masks during use. A few recent studies have assessed worst-case scenarios, examining the leaching of Ag from masks in artificial saliva or other leaching solutions. Leaching experiments in artificial saliva by Montalvo et al. (2023) showed that Ag was released in amounts varying from 0.03 up to 36% of total Ag content, in four out of eight Ag-containing face masks. Pollard et al. (2021) reported that leaching varied widely across manufacturer, particle type, and solution, and reached as much as 100% of Ag content after 8 h of exposure. Dermal exposure associated with the use of Ag-coated face masks estimated using skin absorption models was found not to exceed systemic no effect levels (Estevan et al. 2022). These studies, however, have consistently agreed that a major limitation is the lack of release data in realistic usage conditions. Although it has been shown that fiber fragments may be released (Kisielinski et al. 2024), there are no data on particle release during inhalation from face masks available in the literature.

As a result, reliable safety assessments of face masks are currently based on the safe-by-design principle. Combining electron microscopy (EM) and inductively coupled plasma mass spectrometry (ICP-MS) allows for the assessment of the intrinsic safety of face masks based on the amount, type, and localization of a specific compound and a comparison with no observed adverse effect levels (NOAEL) derived from data for inhalation by the National Institute for Occupational Safety and Health. This approach was first introduced for TiO_2, a suspected human carcinogen (ECHA RAC 2017), after it was observed that aggregates/agglomerates (AA) of TiO_2 particles were observed in all types of synthetic textile fibers in examined face masks, despite not being explicitly advertised (Verleysen et al. 2022). These TiO_2 AA, referred to as textile-grade TiO_2, have specific size properties to provide optimal absorption, giving a whitening and matting effect to the mask. Later, the method was adapted for Ag (Mast et al. 2023), making a distinction between Ag NPs and Ag^{+} ions and has been proposed as a method for evaluating compliance with the existing EU regulations concerning approval, labeling and safety of use (Mast et al. 2023).

As a diverse range of Ag-based biocides have been reported in the literature, this study aims to contribute to the refinement of specifications of face masks safe for consumer use, and to evaluate whether currently applied safe-by-design assessments should be adapted. Hereto, types of particulate Ag biocides actively applied to face masks available on the market were identified, the localization of biocides within or as a coating on the fibers was assessed and their elemental composition was determined using scanning transmission electron microscopy coupled with energy-dispersive X-ray spectroscopy (STEM-EDX).

Materials & methods

Mask selection

Ten face masks including cloth masks (intended for use by the general public), surgical masks, and respirators (types illustrated in Figure 1 in Everaert et al. (2025)), mostly originating from the French market, were selected for characterization and labeled M1 to M10 (Table 1). Only M8, a prototype for an innovative ultraviolet (UV)-integrated photocatalytic mask, to which the authors' were provided access for the purposes of this study, was not yet available on the market. Seven out of 10 masks were advertised to contain a form of Ag biocide, one of the masks' advertisements mentioned the use of a biocide consisting of NPs without specific mention of the type of biocide, and two masks had no type of biocide advertised and were selected as negative controls. For the respirator mask type, only a negative control FFP2 mask could be included since no respirator containing Ag was found on the market.

Sample preparation

Since masks usually consist of multiple layers of textile, the layer with the highest Ag content was identified using ICP-MS as described in Ouf et al. (2024) and selected for subsequent STEM-EDX analysis. Mask layers were numbered starting with the inner layer (closest to the mouth) and ending with the external layer (furthest from the mouth). In masks M6, M9, and M10, where the Ag content of each layer was below the detection limit, the layer with highest titanium (Ti) content, as determined by inductively coupled plasma assisted optical emission spectroscopy (ICP-OES) (Ouf et al. 2024), was selected for STEM-EDX analysis. The preparation of TEM specimens from the mask layers followed the methodology described by Wouters et al. (2022) and as applied by Verleysen et al. (2022) and Mast et al. (2023). Briefly, ultra-thin sections of the textiles were prepared by embedding the textile piece in epoxy resin, followed by ultra-thin sectioning using an ultramicrotome (Leica Microsystems, Vienna, Austria). TEM specimens were prepared from three replicate masks per mask type.

Imaging and analysis

High angle annular dark field (HAADF)-STEM-EDX analysis was performed using a Talos F200S transmission electron microscope equipped with a HAADF detector and a Super-X EDX detector (Thermo Fisher Scientific, Eindhoven, The Netherlands). Imaging was done using Velox software (V3.11, Thermo Fisher Scientific). The sizes of constituent particles (CP) of Ag and other particulate components, when present, were measured manually, and expressed either as the median minimum Feret diameter (F_{min}) or as a range of F_{min} values when too few particles were observed to produce meaningful statistics. For textile-grade TiO_2, when observed, AA sizes were determined and expressed as the median F_{min}. This analysis was done semi-automatically using the ParticleSizer plugin in ImageJ, as described in Verleysen et al. (2022). The size measurement of particles was done on images that were taken before acquisition of the EDX scan to avoid possible beam damage and carbon contamination issues.

Results

Ultramicrotomy combined with STEM-EDX analysis allowed identifying different types of Ag-based biocides and determining their localization embedded within fibers (Figure 1) or on the surface of fibers (Figure 2). Additional illustrative images, EDX data and spectra are provided in Supplementary Information A (SA).

In M1, the biocide was present under the form of a Ag-synthetic amorphous silica (SAS) composite, which was embedded in the fibers (Figure 1a). In high magnification HAADF-STEM images, the composite appeared as high intensity Ag NPs surrounded by less

Table 1. Overview of the characterized mask samples with description of the type of biocide applied.

Mask	Type	Layer selected	Composition	Type of biocide advertised
M1	Cloth	1 of 3	Polyamide	"Silver-silica composite that contains nanosilver sintered onto amorphous silicon dioxide"
M2	Cloth	3 of 3	85% polyester + 15% elastane	"Reaction mass of titanium dioxide and silver chloride"
M3	Surgical	1 of 3	Polypropylene	"Silver chloride"
M4	Surgical	3 of 4	Silver fabric antibacterial paper	"Nano silver"
M5	Cloth	3 of 3	50% cotton + 50% polyester	"Nano silver ions"
M6	Cloth	1 of 3	80% polyester + 20% elastane	"Nanoparticles"
M7	Cloth	3 of 3	Cotton	"Viral stop Ag (silver chloride)"
M8	Cloth (prototype)	1 of 3	PET + optical fiber	"Silica (W7622) and TiO_2 (P25) coated Ag wire"
M9	FFP2 respirator	4 of 5	Polypropylene	No biocide advertised – negative control
M10	Surgical	1 of 3	Polypropylene	No biocide advertised – negative control

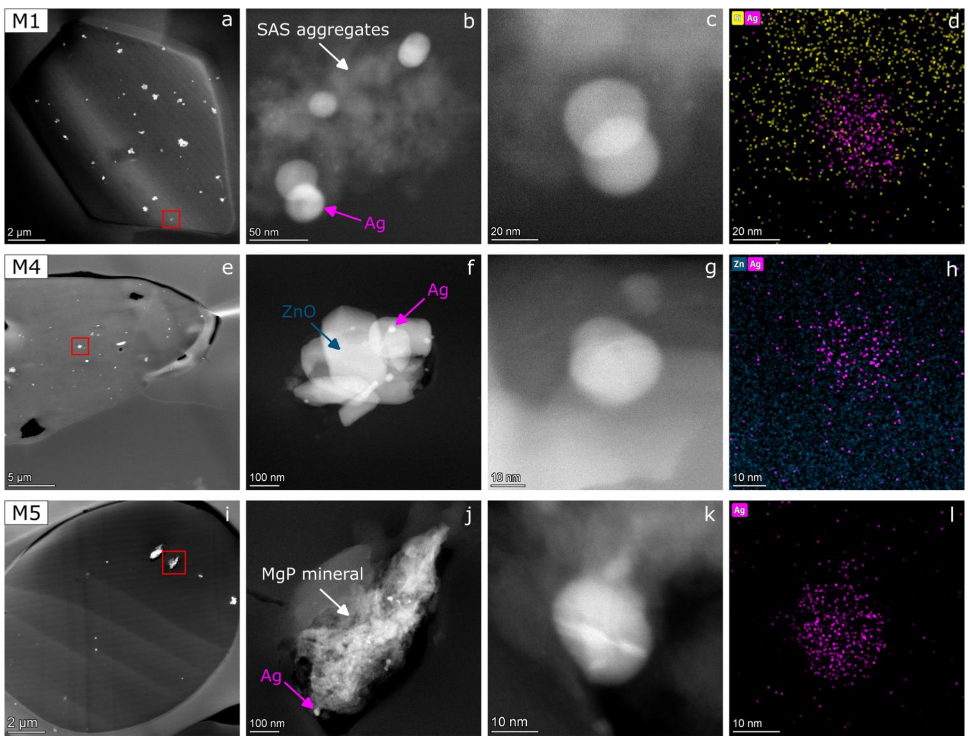

Figure 1. STEM-EDX analysis displaying the type, localization and composition of Ag-based biocides embedded within fibers of masks M1, M4, M5. Low magnification HAADF-STEM images of the fiber cross sections with high intensity particles (a,e,i). HAADF-STEM images of the biocide in the area indicated by the red square (b,f,j). High magnification HAADF-STEM images of the silver particle indicated by a magenta arrow (c,g,k). EDX spectral images of observed elements (Ag: magenta; Si: yellow; Zn: dark blue) corresponding to the neighboring HAADF image (d,h,l). Extended STEM-EDX data are presented as Supplementary Information A.

intense SAS aggregates (Figure 1b–d, Figures SA1 and SA2) (Rasmussen et al. 2013). Some isolated Ag NPs were embedded in fibers as well (Figure SA3). The size of Ag NPs in this mask varied from approximately 3 to 55 nm.

In M2, part of the sectioned fibers held particles coated on the surface (Figure 2a). The coated particles consisted of a mix of TiO_2 particles and few relatively small Ag and AgCl particles embedded within a silicon (Si)-containing background (Figure 2b–d, Figures SA4 and SA5). Ag and AgCl particles ranged in size from approximately 10 to 200 nm, and TiO_2 particles on the surface of fibers ranged in size from approximately 50 to 380 nm. The Si-containing area seemed

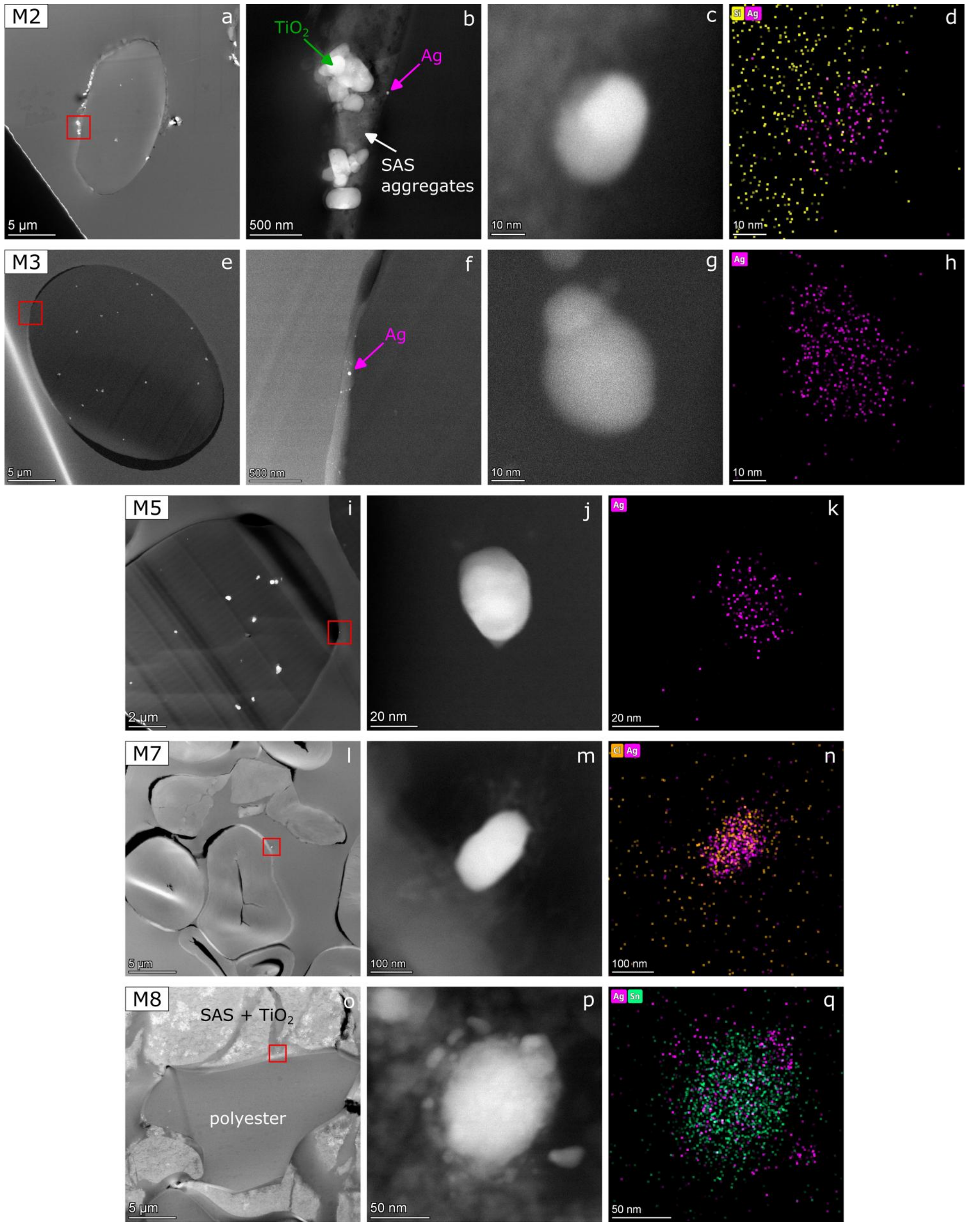

Figure 2. STEM-EDX analysis displaying the type, localization and composition of Ag-based biocides on the surface of fibers of masks M2, M3, M5, M7, and M8. Low magnification HAADF-STEM images of the fiber cross sections with high intensity particles (a,e,i,l,o). HAADF-STEM images of the biocide in the area indicated by the red square (b,f,j,m,p). High magnification HAADF-STEM images of the Ag particle indicated by a magenta arrow (c,g). EDX spectral images of observed elements (Ag: magenta; Si: yellow; Cl: orange; Sn: green) corresponding to the neighboring HAADF image (d,h,k,n,q). Extended STEM-EDX data are presented as Supplementary Information A.

to consist of SAS aggregates, similar to those in M1, as indicated by the HAADF intensity contrast (Figure SA5h).

In M3, Ag NPs with sizes ranging from 7 to 35 nm were observed on the surface of polypropylene fibers (Figure 2e–h). While the majority of Ag NPs appeared as isolated particles, in the case shown here a group of NPs were found in close proximity to each other. For some particles, EDX data showed a small Cl and/or S peak in addition to Ag (Figure SA6).

In M4, fibers contained isolated Ca-containing particles (probably $CaCO_3$) and both isolated and aggregated/agglomerated Zn-containing particles (probably ZnO) (Figure 1e, Figure SA7). Some Zn-containing aggregates/agglomerates had small Ag NPs attached, with sizes ranging from 10 to 30 nm (Figure 1f–h, Figure SA8). EDX analysis indicated that the Ag NPs were pure.

In M5, cotton fibers were biocide-free, while within polyester fibers micrometer-sized mineral complexes were observed (Figure 1i). The main peaks in the

EDX spectrum of the minerals correspond to Mg, P, and O, indicating the presence of magnesium phosphate (hereafter referred to as MgP), while smaller peaks corresponding to Si and Al indicated the presence of these elements (Figure SA9). Within the MgP mineral, NPs with high HAADF intensity were observed, demonstrated to consist of Ag by EDX analysis (Figure 1j-l, Figure SA9). These Ag NPs were especially beam-sensitive in this mask: when imaged at high magnification, increasing the beam intensity per area, the Ag particle transformed its shape, which made accurate assessment of the original size/shape problematic. Therefore, to avoid possible artificial influence of the electron beam on particle size, size estimation was done before the EDX scan, which required a long irradiation time. Apart from the Ag within MgP, isolated Ag NPs and small agglomerates of Ag NPs were observed within fibers and on the surface of fibers (Figure 2i–k, Figure SA10). The size of Ag NPs in M5 varied from approximately 5 to 35 nm.

Although in M6 no Ag particles were detected and no Ag signal was measured in the EDX spectra of analyzed areas, some fibers were partially coated with particles on their surface, as shown in Figure 3a. The aggregated and agglomerated particles on the surface exhibited a Si signal that is indicative for silica (Figure 3b and c, Figure SA11). The morphology of the aggregates/agglomerates and the estimated constituent particle size of 10–100 nanometer are similar to that of SAS (Rasmussen et al. 2013).

In M7, AgCl micro- and nanoparticles were identified on the surface of cotton fibers (Figure 2l–n, Figure SA12). In one of the examined replicates, no particles were detected, indicating that the deposition of AgCl particles on the surface of fibers was not homogeneous throughout the mask. The particles were isolated and their size ranged from 50 to 250 nm.

M8 consisted of polyester fibers mixed with other μm-sized structures whose cross-sections displayed various shapes and sizes (Figure 2o). Ag-containing NPs, with sizes ranging from 20 to 100 nm were observed on the surface of the polyester fibers. These NPs consist of Ag only (Figure SA13), a mixture of Sn and Ag (Figure 2p and q, Figure SA14), or a mixture of Ag and S (probably Ag_2S, Figure SA15). The μm-sized structures surrounding the polyester fibers consisted of a mix of two types of aggregated/agglomerated NPs with different intensities in the HAADF images. EDX analysis showed that the low intensity particles displayed an Si signal indicative of SAS and the high intensity particles displayed a Ti signal indicative of TiO_2 (Figure 3d–g, Figure SA16). TiO_2 agglomerates were also present as a direct coating on polyester fibers (Figure 3h–j) in some cases. Although their borders were not always easy to discern due to the overlap of many TiO_2 particles, their constituent particle sizes were estimated to be in the range of 10 to 50 nm.

For M9 and M10, and in agreement with the ICP-MS measurements previously conducted by Ouf et al. (2024), no Ag particles were identified. This demonstration of the absence of Ag in ionic or particulate form confirmed the choice of masks M9 and M10 as negative control samples regarding the use of Ag as a biocide.

In addition to the previously described particles/compounds, TiO_2 AAs were present within all synthetic fibers of the analyzed masks, with physicochemical properties matching that of so-called textile-grade grade TiO_2, as described in Verleysen et al. (2022) (Supplementary Information B, Figure SB1). The TiO_2 AA had median aggregate/agglomerate sizes ranging from 129 to 214 nm across different masks.

Discussion

Given that the risks linked to prolonged exposure to Ag-functionalized face masks are still unknown, it is important to assess whether currently applied safe-by-design assessments are adequate or should be adapted. To that end, a selection of Ag-advertised face masks were characterized to evaluate the types of particulate Ag-based biocides present in commercial face masks.

Diversity and function of silver-based and other biocides

The literature search and the experimental data (summarized in Table 2) illustrated that several types of Ag biocides, marketed specifically for anti-microbial properties, were applied to face masks available on the market at the time of purchase in 2023. Most observed types aligned well with the advertised form of the Ag biocide (Table 1). However, in some cases, certain components of the biocide formulation were identified that were not mentioned on the labels, including SAS, ZnO, MgP, and Sn in masks M2, M4, M5, and M8, respectively.

The main types of Ag biocides that were reported in literature for use in textiles were shown to be represented in the face masks: metallic Ag in the form of isolated Ag NPs or Ag NPs as part of composite

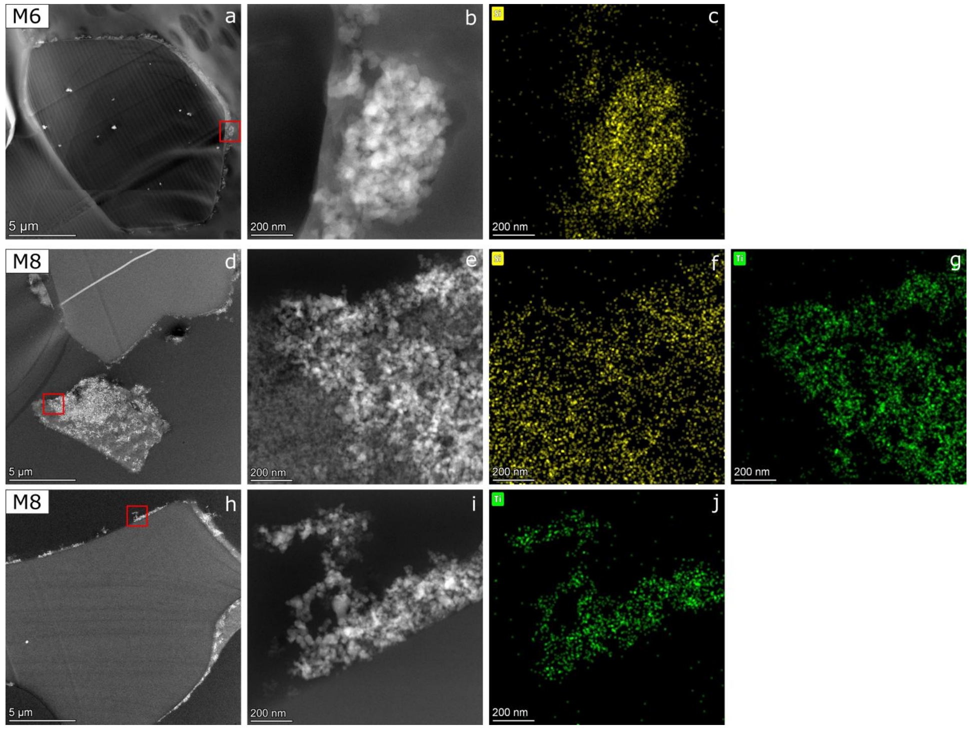

Figure 3. STEM-EDX analysis displaying the type, localization and composition of other surface-coated biocides in masks M6 and M8. Low magnification HAADF-STEM images of the cross sections of polyester fiber (a), SAS+TiO_2 structure (d) and PET fiber (g) with high intensity particles. High magnification HAADF-STEM images of the biocide in the area indicated by the red square (b,e,i). EDX spectral images of observed elements (Si: yellow; Ti: green) corresponding to the neighboring HAADF image (c,f,g,j).

structures, Ag salts in the form of AgCl NPs and Ag_2S NPs, and Ag ion exchangers in the form of Ag-embedded/coated MgP. A few observed formulations have not yet been documented in the literature. While ZnO NPs have been demonstrated as an antimicrobial compound in surgical masks (Qadir et al. 2024), the Ag NP – ZnO composite, to the authors' knowledge, has not been described in the literature. Similarly, although combinations of Ag with silica (Brzezinski et al. 2007; Balagna et al. 2020) or TiO_2 (Li et al. 2006; Chacon-Argaez et al. 2023; İbili and Daşdemir 2023) are known, the triple composite Ag(Cl) NP – TiO_2 – SAS remains unreported. Additionally, the Ag NP – Sn composite is not yet reported.

A notable prevalence of AgNP composites or compounds was observed, which can serve various purposes. First, to reduce the risk of emergence of Ag-resistant micro-organisms, a second antimicrobial compound—often a metaloxide such as TiO_2 or ZnO—can be incorporated (Li et al. 2006). These metal oxides provide biocidal action through the generation of reactive oxygen species (ROS) (Li et al. 2006; Raghupathi et al. 2011). This mechanism of action was also applied in M8, where TiO_2 particles embedded on SAS are supposed to be photocatalytically activated through an integrated UV light source to enhance ROS production and biocidal activity in addition to the biocidal effect of Ag, as claimed by the manufacturer. It should be noted that the physicochemical properties of the catalytic TiO_2 specifically used in M8 and of the textile-grade TiO_2 incorporated into all synthetic mask textiles are different. Textile-grade TiO_2, with median CP F_{min} size in the range of 97 to 197 nm and median AA F_{min} size in the range of 129 to 214 nm, are engineered to achieve high refractive power (Peters et al. 2014), which contributes to the whitening, matting, and improved stability of the textile material. In contrast, P25 TiO_2, commonly utilized for photocatalytic applications, consists of smaller CP sizes ($\sim$10-50 nm), maximizing the surface-to-volume ratio to enhance its catalytic and thus biocidal efficiency (Karthikeyan Thirunavukkarasu et al. 2022).

Secondly, the formation of an Ag-incorporated compound might facilitate controlled Ag release. The slow and controlled release of Ag^+ ions from a Ag NP – SAS nanocomposite, as applied in the formulation in M1, is described by Egger et al. (2009). Unlike the more typical application of Ag NP – SAS composites as sputtered coating (Balagna et al. 2020), in mask M1, the biocide was embedded within the fibers, which could further enhance a gradual Ag^+ release. A similar strategy seemed to have been applied in M2, where both Ag/AgCl and TiO_2 particles were embedded within a SAS matrix applied as a coating to

Table 2. Summary of the results of the STEM-EDX analyses of particles in face masks and their applications.

Mask	Type of particles	Localization	CP size[a] (nm)	AA size[b] (nm)	Reported application in textiles/other
M1	Ag NP – SAS composite	Embedded	3–35 (Ag)	ND[c]	Virucidal silver nanocluster/silica composite sputter coating on FFP3 mask (Balagna et al. 2020) Bacteriostatic polymeric coating material for textiles (Brzezinski et al. 2007)
	Ag NP	Embedded	10–55	NA[d]	Biocidal action of Ag NP in various textiles (Zhang et al. 2009; Montes-Hernandez et al. 2021; Abazari et al. 2023)
	TiO_2 AA	Embedded	97	147	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M2	Ag(Cl) NP – TiO_2 – SAS composite	On surface	10–200 (Ag)	ND[c]	Improved photocatalytic and biocidal activity of Ag–TiO_2 composites on cotton (Chacon-Argaez et al. 2023) AgCl-TiO_2/dendrimer-based nanoparticles for superhydrophobic and antibacterial textiles (İbili and Daşdemir 2023)
	TiO_2 AA	Embedded	135	166	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M3	Ag NP	On surface	7–35	NA[d]	Biocidal action of Ag NP in various textiles (Zhang et al. 2009; Montes-Hernandez et al. 2021; Abazari et al. 2023)
	TiO_2 AA	Embedded	157	176	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M4	Ag NP – ZnO composite	Embedded	10–30 (Ag)	ND[c]	Enhanced antimicrobial properties of ZnO NP coated cotton (Giedraitienė et al. 2024)
	$CaCO_3$	Embedded	ND[c]	ND[c]	Durability, whiteness, mattening (Brunner et al. 2021)
M5	Ag-embedded and coated MgP	Embedded	5–35 (Ag)	ND[c]	Ag-doped magnesium phosphate microplatelets as antibacterial orthopedic biomaterials (Sikder et al. 2020)
	Ag NP	Embedded + On surface	20–35	NA[d]	Biocidal action of Ag NP in various textiles (Zhang et al. 2009; Montes-Hernandez et al. 2021; Abazari et al. 2023)
	TiO_2 AA	Embedded	132	143	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M6	SAS aggregates	On surface	20–100	ND[c]	Superhydrophobicity (Xue et al. 2009; Borah et al. 2023)
	TiO_2 AA	Embedded	111	160	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M7	AgCl NP	On surface	50–250	NA[d]	Antibacterial AgCl NP coating (Potiyaraj et al. 2007; Abbasi et al. 2011)
	TiO_2 AA	Embedded	107	164	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M8	Ag NP	On surface	20–30	NA[d]	Biocidal action of Ag NP in various textiles (Zhang et al. 2009; Montes-Hernandez et al. 2021; Abazari et al. 2023)
	Ag-Sn NP	On surface	75–100	NA[d]	Not reported
	Ag_2S NP	On surface	25–65	NA[d]	Not reported
	TiO_2 AA + SAS aggregates	On surface	10–50 (TiO2)	ND[c]	Photocatalyzed biocidal action and self-cleaning (Rashid et al. 2021)
	TiO_2 AA	Embedded	98	129	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M9	TiO_2 AA	Embedded	167	171	Whiteness, opacity, UV stability (Verleysen et al. 2022)
M10	TiO_2 AA	Embedded	179	214	Whiteness, opacity, UV stability (Verleysen et al. 2022)

[a]Constituent particle size expressed as the median F_{min} or as a range of F_{min} values, measured or estimated manually.
[b]Aggregate/agglomerate size expressed as the median F_{min} determined semi-automatically using ImageJ.
[c]ND = Not determined.
[d]NA = Not applicable.

fibers, which allowed the effective antimicrobial properties of textiles to be sustained over a long time, even after several washings (Tomšič et al. 2009). The inclusion of Ag in MgP, known for its application as a next-generation antibacterial material in orthopedic contexts (Sikder et al. 2020), also results in a sustained Ag^+ release. These findings illustrate the diverse strategies employed in mask design to apply silver's antimicrobial properties.

In mask M6, SAS was present as a coating on the fibers. While SAS is not intrinsically biocidal, textiles coated with silica NPs have been reported to exhibit superhydrophobic properties (Xue et al. 2009; Borah et al. 2023), which may act as a strategy to fight virus transmission. In mask M4, $CaCO_3$ particles were incorporated to enhance durability, whiteness, and matting effects, and can serve as an alternative to TiO_2 (Brunner et al. 2021).

Implications for risk assessment

Given the variety of Ag-based (and other) biocide types applied in face masks, it is important to consider their implications for risk assessment. In the absence of reliable data on Ag release from these materials in realistic usage conditions, risk assessments can be guided by a safe-by-design approach, as introduced by Verleysen et al. (2022) and Mast et al. (2023). Their method allows to identify masks that are safe-by-design based on the amount of Ti and Ag measured by ICP-MS and on the localization of the NPs, fiber diameters, and the NOAELs of Ag^+ ions and Ag NPs, and those of TiO_2 NPs. These methods can, in principle, also be applied for the above-mentioned biocidal compounds provided that the NOAEL of each compound is known. Importantly, in masks M1, M2, M4 and M8, the presence of a combination of several compounds was observed. In those cases, co-exposure/combined effects must be considered, as indicated by Forest (2021). Also, the combination of Ag NPs and ions could influence toxicity (Lai et al. 2020). Accordingly, detailed physicochemical characterization and substance-specific risk assessments are required to evaluate the safety of face masks containing complex biocidal formulations.

In the majority of masks containing Ag (5 out of 7), Ag-containing NPs were identified on the surface of fibers. Additionally, significant amounts of other nanoparticulate materials (SAS, TiO_2) were found coating the fiber surfaces. As with Ag, both SAS and TiO_2 raise potential toxicological concerns. After the Risk Assessment Committee (RAC) of the European Chemical Agency (ECHA) reviewed the carcinogenic potential of TiO_2 (ECHA RAC 2017), it was classified as suspected human carcinogen by inhalation. In the case of SAS, studies show that particle size, composition, synthesis and processing methods can affect its toxicity (Croissant et al. 2020). In half of the masks labeled as containing a biocide, Ag or other biocidal materials were applied to the inner layer, where the potential for inhalation exposure is likely the highest. These observations highlight the need to develop methods for assessing the release of Ag and other applied substances, in any form, from the fiber surfaces to support further exposure and risk assessments.

The methods outlined in this publication provide a solid foundation for initiating Ag NP release studies and exposure assessments. ICP-MS/OES combined with STEM-EDX allows evaluation of total Ag content, determination of its presence in NP form, and localization of Ag NP and other NP either within or on the surface of fibers, all key information for risk assessment. This information can guide further tests to evaluate the release of Ag under realistic conditions such as through a simulated breathing setup that directs and controls airflow through the mask and captures airborne particles on filters or grids. It can be anticipated that high-end analytical techniques need to be applied due to the low mass of Ag present in masks. Total released Ag or other compounds collected on the filter can be quantified by ICP-MS/OES, while size and elemental composition of particles captured on the grid can be estimated using STEM-EDX.

Limitations

The methodology, based on ultramicrotomy and STEM-EDX as applied in previous studies (Verleysen et al. 2022; Wouters et al. 2022; Mast et al. 2023), allowed localization (inside or on the fiber), and identification (isolated or as compound or composite) of biocides in various face masks of different textile types.

One of the main limitations of the applied methodology is its limited sampling, since the cross-sectional samples are 100–150 nm in thickness and the studied area in TEM is limited to approximately 1–2 mm^2 per grid. However, the results for biocide detection and identification were generally reproducible across different mask replicates, indicating that the applied sampling method was sufficient for detection purposes. Despite this, the low detection frequency of Ag-based biocides, and consequently the low number of Ag particles detected, limited the ability to quantitatively assess the size of Ag particles with statistical relevance. Additionally, the ability to interpret the dispersity of the biocide, an important factor for mask performance, by image analysis parameters like the interparticulate distance, was limited. To analyze larger sample volumes and assess dispersity of particles on fiber surfaces, scanning electron microscopy (SEM) could serve as an alternative or complementary technique. A limitation of SEM, however, lies in the lower resolving power of SEM compared to TEM, which may result in the inability to detect Ag particles smaller than 10 nm (Boyd et al. 2011). For biocide dispersity within fibers, techniques such as X-ray nanotomography and time-of-flight secondary ion mass spectrometry can provide larger-volume and 3D-resolved information, but their limited spatial resolution may hinder the detection and precise localization of individual NPs.

While STEM-EDX enables spatially resolved elemental analysis, there are inherent limitations in definitively identifying specific compounds. These limitations arise from background signals in the EDX spectra, including carbon from the supporting film, oxygen from ambient contamination, and silicon and copper signals resulting from X-ray fluorescence induced in the detector and grid components by primary X-rays emitted from the sample. Such background contributions may overlap with signals from the sample, making it challenging to clearly determine the chemical compounds present in low quantities. Moreover, accurate quantification of elemental percentages is generally not feasible under these conditions. However, the presence of elements such as silicon in the sample can still be qualitatively confirmed by comparing the intensity of characteristic peaks in the EDX spectrum with those from a background region (e.g., Figure SA4).

The sensitivity of the EDX measurement is limited by the selected magnification. In STEM images acquired at a magnification where both small (<50 nm) Ag particles and larger components, such as TiO_2 and SAS, were present, the Ag signal was often non-detectable in the corresponding EDX spectrum (e.g., $7.16 \cdot 10^5$ x; Figure SA2d). Reliable identification of such small Ag particles requires higher magnification (e.g., $> 4 \cdot 10^6$ x; Figure SA2f) to ensure sufficient signal-to-noise ratio for elemental detection. The sensitivity and accuracy of the EDX measurement also depends on the depth of the particle of interest within the section. If the particle is located at the bottom of a 100 to 150 nm thick resin-embedded section, the electron beam may already be significantly broadened due to scattering. This reduces the probe current density at the intended scanning point and can result in weaker excitation of X-rays from that location and additional excitations from surroundings.

Conclusions

Using STEM-EDX, a variety of Ag-based biocides, including forms of metallic Ag, Ag salts in NP form, Ag ion exchangers, and numerous composite structures, were identified in face masks. These composites can combine different modes of biocidal action or facilitate gradual Ag release, thereby extending antimicrobial activity, and improved resistance to Ag NP loss due to washing. The Ag-based biocides observed in face masks were similar to those reported for textiles in general. These findings have implications for safety and risk assessment. Safe-by-design approaches must account for the NOAEL of all incorporated compounds. Furthermore, the effects of co-exposure to multiple NPs and their additive or synergistic effects should be addressed. The detected presence of various NPs, including Ag, TiO_2, and SAS, on fiber surfaces highlights the need for release studies under realistic usage conditions to evaluate potential release and consumer exposures. The developed methodology can help support implementing regulations, such as the Biocidal Products Regulation (Regulation No 528/2012 of the European Parliament and of the Council of 22 May 2012 Concerning the Making Available on the Market and Use of Biocidal Products 2012). It enables verification of whether biocides align with the advertised descriptions or labeling and facilitates the identification of certain biocidal components which are not mentioned on the labels. The findings of this study can contribute to regulatory control of the quality of face masks, emphasizing the importance of standardized analytical methods to ensure compliance and protect public health.

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Disclosure statement

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Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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